IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 5 July 2026, accepted 28 July 2026, date of publication 12 August 2026, date of current version 19 August 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3722999

RESEARCH ARTICLE

A Task-Aware Compression Framework for Efficient Transmission of Marine Mammal Embeddings in Bandwidth-Constrained Acoustic Networks

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This work was supported by Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, India.

ABSTRACT Real-time passive acoustic monitoring is fundamentally limited by the constraints of underwater telemetry. Acoustic modems typically provide bandwidths of only tens to hundreds of bits per second, which are insufficient for transmitting the high-dimensional acoustic data generated by monitoring systems in real time. Modern audio foundation models achieve state-of-the-art classification accuracy but generate high-dimensional embeddings that are operationally impractical to transmit from edge deployments. This paper addresses the critical trade-off between accuracy and bandwidth, proposing a method to enable verifiable, high-performance monitoring within strict transmission budgets. We propose a task-aware compression pipeline evaluated on a curated 15-class subset of the Watkins Marine Mammal Sound Database. We benchmark three pre-trained audio embedding models, identifying a bioacoustic foundation model as the superior but overparameterized feature extractor. We employ a supervised bottleneck autoencoder to distill task-discriminative features with varying loss functions, followed by product quantization and optimized product quantization to minimize the bit-rate while preserving class separability. Our approach achieves extreme compression ratios of 256 to 768 times while retaining over 97 percent of the initial baseline classification accuracy. Experimental results demonstrate that classification on the compressed latent representation matches or surpasses full-embedding performance. Information-plane analysis attributes this to the removal of recording-condition artifacts that caused mild overfitting in the uncompressed baseline. These findings demonstrate that the generalization tax of large foundation models can be dramatically reduced for domain-specific applications. The proposed framework enables practical marine monitoring systems that combine the accuracy of foundation models with the efficiency required for underwater acoustic networks.

INDEX TERMS Bio-acoustics, passive acoustic monitoring, supervised autoencoder, task-aware compression.

I. INTRODUCTION

Passive Acoustic Monitoring (PAM) represents a foundational methodology in marine bioacoustics, utilizing specialized underwater transducers to record and analyze biological soundscapes. As a non-invasive remote sensing technique, PAM is essential for marine mammal conservation, facilitating the high-resolution detection, localization, and classification of species-specific vocalizations. A standard PAM architecture integrates a modular suite of hardware and software components, typically comprising hydrophone arrays for signal acquisition, robust deployment platforms, and specialized post-processing frameworks for automated data analysis [1], [2], [3], [4]. Real-world deployment in remote oceanic environments is fundamentally limited by a critical acoustic bandwidth bottleneck. Although commercial

The associate editor coordinating the review of this manuscript and approving it for publication was Yougan Chen.

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acoustic modems cite peak data rates between 15 and 30 kbps, such throughput is typically restricted to idealized vertical links or short-range conditions. In long-range, shallow-water deployments, signal integrity is degraded by significant multipath reverberation and Doppler spread. To maintain link stability under these stochastic conditions, systems must utilize robust, incoherent modulation schemes such as Frequency Shift Keying (FSK) or spread spectrum rather than high-rate coherent methods. Consequently, the effective reliable telemetry throughput often diminishes to a range of 64–640 bps [5], [6] as detailed in Table 1. Engineering for restricted bandwidth is therefore a fundamental requirement for operational reliability rather than a conservative design precaution.

Underwater acoustic modems operating in the 64–640 bps range impose severe transmission limitations, rendering conventional approaches impractical. A typical high-dimensional audio embedding (1536–2048 dimensions at 32 bits per value) requires 48–256 kilobits of storage and transmission bandwidth per clip, which is unaffordable for systems monitoring continuously for weeks or months on battery-powered platforms.

TABLE 1. Overview of commercially available acoustic modems suitable for PAM telemetry. The Bitrate column reflects robust operating modes rather than theoretical maximums. Costs are approximate market values as of 2026.

Manufacturer	Model	Cost (USD)	Bitrate (bps)	Range (m)
Water Linked	Modem M64	∼2,300	64	200
Desert Star	SAM-1	∼1,800	100	1,000
Popoto Modem	Popoto OEM	∼2,750	80–5,120	3,000+
Tritech	Micron Modem	∼4,000	40	500
Blueprint Subsea	SeaTrac X150	∼5,500	100	1,000
DSPComm	AquaComm Gen2	∼4,500	100–480	3,000
Teledyne Benthos	ATM-900 Series	>8,000	140–2,400*	2,000–6,000
EvoLogics	S2C Series	>10,000	500–31,200*	1,000–8,000

*Higher rates achievable only in vertical/short-range links; standard long-range rates are lower.

Recent advances in audio foundation models—such as *Perch* [7], a specialized bioacoustic foundation model pre-trained on a large corpus of wildlife recordings—have achieved unprecedented classification accuracy. These models produce semantically rich but overparameterized embeddings, as they were not trained for the specific task of marine mammal classification. For domain-specific tasks (e.g., 15-class marine mammal classification), much of this representational capacity is spent on task-irrelevant features [8]. This phenomenon, termed the *generalization tax* [8], implies that naive transmission of full embeddings is inefficient: the model encodes information irrelevant to classification but necessary for broad generalization.

A natural alternative to transmitting embeddings is *on-edge recognition*, in which classification is performed entirely at the edge sensor and only the predicted label or a detection alert is transmitted. Both paradigms require running the foundation model at the edge to extract features; the key distinction lies not in computational cost but in what is transmitted and what downstream analyses remain possible. On-edge recognition minimizes the payload for a fixed, closed-set task, yet the transmitted decision is unfalsifiable: a remote analyst cannot audit a detection, re-score it using an updated model, or apply it to tasks the device was not originally configured for. Transmitting a compact, task-aware embedding instead preserves verifiability by allowing past detections to be re-examined, enables central model evolution by permitting downstream classifiers to be retrained or extended to new species without the prohibitive expense of physically recovering moored underwater hardware, and supports the multi-task reuse of a single transmitted vector. Our framework occupies a middle ground between these approaches, preserving a generalization-capable representation while compressing it sufficiently to operate within a strict bandwidth budget, as summarized in Table 2.

TABLE 2. Comparison of remote-monitoring paradigms under bandwidth-constrained underwater telemetry.

	On-edge recognition	Full embedding	Task-aware (proposed)
Payload per clip	∼4 bits	48–256 kbit	8–16 bytes
On-device compute	Perch + classifier	Perch only	Perch + bottleneck
Modem TX duration/clip	<1 ms	>100 s (infeasible)	≤1 s
Verifiable downstream	No	Yes	Yes
Re-trainable server-side	No	Yes	Yes (within task)
Open-set / new tasks	Requires reflash	Yes	Limited

This work directly addresses the accuracy–bandwidth trade-off in marine PAM by proposing a *task-aware compression pipeline* that eliminates semantic redundancy. We evaluate three embedding families—VGGish [9], PaSST [10], and Perch [7]—and implement a supervised bottleneck autoencoder that jointly optimizes reconstruction and classification objectives. By further applying Product Quantization (PQ) [11] and Optimized Product Quantization (OPQ) [12], we achieve extreme compression ratios (256× to 768×) while retaining over 97% of baseline classification accuracy.

The key insight is that *compression and classification are not separate problems*: by explicitly optimizing for the downstream task, we discard task-irrelevant information that unsupervised methods—such as standard Autoencoders [13] or PCA—would attempt to preserve. This enables verifiable, reproducible compression suitable for mission-critical conservation applications.

A. CONTRIBUTIONS

1) A practical task-aware, constraint-driven embedding compression pipeline demonstrating 256–768× compression with over 97% F1 retention.
2) A comparative evaluation of three foundation models (VGGish, PaSST, Perch) on a curated 15-class marine mammal dataset, revealing Perch's superior out-of-distribution generalization despite its higher dimensionality.
3) A supervised bottleneck autoencoder that jointly optimizes reconstruction (MSE) and classification (cross-entropy) losses, enabling compression-aware feature learning.

4) A quantization study (PQ vs. OPQ) demonstrating that extreme compression (8 bytes per embedding, suitable for 64 bps modems) is achievable with over 97% accuracy retention.
5) Empirical validation that direct classification on compressed latent representations matches full-embedding performance, indicating that discarded information is genuinely redundant for the task.

II. LITERATURE REVIEW

This section provides an overview of the architectures and models currently used to address the challenges of the Watkins Marine Mammal Sound Database (WMMD), as well as the data transfer bottlenecks inherent in underwater telemetry. To systematically explore these overlapping domains, this review is divided into several key areas. First, we examine the state-of-the-art in marine mammal acoustic classification to establish current benchmarks. Next, we evaluate various pre-trained embedding models to understand how different architectures process acoustic features. We then review autoencoder-based compression and vector quantization techniques, which are critical for reducing data size. Finally, we explore transfer learning paradigms and the concept of ''semantic redundancy'' to frame the specific research gaps our proposed task-aware compression pipeline aims to fill.

A. MARINE MAMMAL ACOUSTIC CLASSIFICATION: STATE OF THE ART

1) DATASET AND BENCHMARKING STANDARDS

The **Watkins Marine Mammal Sound Database (WMMD)** [14] has emerged as the definitive benchmark for marine mammal classification research, with recent work shifting toward deep learning-ready subsets. Licciardi and Carbone [15] introduced WhaleNet, a novel deep learning architecture for WMMD species classification, reporting approximately 97.6% accuracy and defining a current state-of-the-art reference point. Current pipelines commonly evaluate on curated subsets spanning 12–15 marine mammal species, typically processed at a standardized 32 kHz sample rate, yielding a 16 kHz Nyquist frequency. This sampling constraint imposes a fundamental limit on the analyzable acoustic range: while sufficient for many odontocetes and mysticetes, vocalizations from species producing very low-frequency infrasonic pulses (e.g., *Balaenoptera musculus*) or ultrasonic echolocation clicks (e.g., *Phocoena phocoena*) may be aliased or attenuated [14], [16].

Beyond WMMD, the field is moving towards continuous monitoring benchmarks. Cominelli et al. [2] highlight the shift from static clip classification to stream-based analysis, a transition that exposes the bandwidth limitations of existing telemetry. This same shift raises the bar for model generalization, as stream-based systems must remain accurate across recording conditions, geographies, and species not explicitly seen during training. To address cross-domain generalization, benchmarks such as **BEANS** [17] have expanded evaluation scopes. By leveraging training corpora spanning thousands of bioacoustic species, models like Perch [7] learn robust, noise-invariant features that can be fine-tuned for the sparse, high-noise environments typical of underwater PAM.

2) REPRESENTATION LEARNING AND INFORMATION BOTTLENECKS

The transition from handcrafted features (e.g., Mel-frequency cepstral coefficients) to deep representation learning has shifted the focus toward large-scale pre-trained models. However, the use of foundation models in PAM is often hindered by the ''generalization tax'' [8]—the retention of acoustic features necessary for broad-scale ecological monitoring but redundant for low-cardinality species identification. This work draws upon the **Information Bottleneck (IB) principle** [18], which posits that an optimal representation should maximize the mutual information with the target label while minimizing the information retained from the input signal. Our supervised autoencoder architecture is a practical implementation of this principle, forcing the model to discard features that do not contribute to species-level classification.

3) PRE-TRAINED EMBEDDING MODELS: COMPARATIVE ANALYSIS

To effectively minimize semantic redundancy, it is first necessary to evaluate the representational capacity of existing foundation models. This subsection provides a comparative analysis of four prominent pre-trained embedding architectures to establish their strengths and domain-specific limitations.

a: VGGISH: GENERAL-PURPOSE AUDIO EMBEDDINGS

VGGish, trained on YouTube-8M with over two million user-labeled audio-video pairs, generates 128-dimensional feature vectors by extracting log-Mel spectrograms over 0.96-second windows. The model's convolutional architecture mirrors VGG image classification networks, applying learnable filters, max-pooling, and fully connected layers to capture local patterns, multi-scale features, and global dependencies. While its shallow architecture enables rapid inference and provides a baseline for terrestrial soundscape datasets, a domain mismatch exists for underwater applications. Because YouTube-8M training heavily features terrestrial speech and music, marine spectral characteristics are largely excluded. VGGish struggles with marine sounds outside the 8 kHz Nyquist limit of its native 16 kHz input sample rate, and when applied to the WMMD its features exhibit poor separation for species such as bottlenose dolphins, beluga whales, and southern right whales.

b: PANNS: HIGH-DIMENSIONAL ACOUSTIC REPRESENTATIONS

The Pre-trained Audio Neural Networks (PANNs) library provides 2048-dimensional embeddings by aggregating

outputs from deep convolutional networks trained on AudioSet [19]. By utilizing a higher-dimensional space, PANNs are designed to capture a wider array of acoustic patterns, which has yielded accuracy improvements over VGGish on specific terrestrial datasets [19]. Although AudioSet pre-training includes diverse sound categories such as animal vocalizations and has been utilized for environmental sound classification (ESC-50 [20]), the increased dimensionality creates a bottleneck for bandwidth-constrained telemetry, and PANNs contain limited explicit marine mammal representation in their training data [19].

c: PASST: TRANSFORMER-BASED AUDIO EMBEDDINGS

The **Patchout fast Spectrogram Transformer (PaSST)** represents a modern alternative to CNN-based embeddings [10]. Built on vision transformer principles adapted to audio spectrograms, PaSST divides log-Mel spectrograms into patches, projects them linearly into embedding sequences, and adds learnable positional encodings, then processes via self-attention layers [10]. Patchout regularization randomly removes frequency bins and time frames during training, reducing overfitting and computational costs. The model outputs 768-dimensional embeddings and achieves state-of-the-art mean average precision (mAP) on multiple audio classification benchmarks [10]. Like VGGish and PANNs, PaSST exhibits domain-mismatch limitations for marine applications.

d: GOOGLE PERCH: FOUNDATION MODELS FOR BIOACOUSTICS

Google's Perch model [7] represents a paradigm shift in audio embeddings, designed specifically for bird and wildlife bioacoustics. Perch is a 12M-parameter EfficientNet-based convolutional model pre-trained on a large, curated corpus of bird and wildlife field recordings spanning thousands of bioacoustic species. Unlike general-purpose models, Perch's training corpus explicitly includes diverse animal vocalizations and ecological contexts [7].

Architecture and Capabilities: Perch processes mel-spectrograms using an EfficientNet-based convolutional encoder, generating both frame-level and clip-level embeddings. The model achieves strong out-of-the-box species-identification performance on held-out bird soundscapes and transfers effectively to novel taxa and geographic regions [7]. Unlike clip-level models (VGGish, PANNs, PaSST), Perch's frame-level design enables fine-grained temporal segmentation, critical for marine mammal calls with complex modulation envelopes.

Marine Applicability: Although trained predominantly on avian and terrestrial data, Perch's architecture and training objectives align well with marine bioacoustic requirements. Frame-level embeddings preserve temporal structure, enabling detection of rapid frequency sweeps (e.g., dolphin whistles) and pulsed calls (e.g., whale clicks). The model's demonstrated generalization to out-of-distribution taxa suggests transferability to marine species not seen during pre-training. However, limited explicit evaluation on marine mammals necessitates careful benchmarking, particularly for low-frequency specialists (fin whales, blue whales) whose calls fall below Perch's pre-training bandwidth [7].

B. AUTOENCODER-BASED COMPRESSION AND FEATURE DISTILLATION

1) VARIATIONAL AUTOENCODERS FOR EMBEDDING COMPRESSION

Variational Autoencoders (VAEs) have emerged as a principled approach for compressing high-dimensional embeddings while maintaining reconstruction fidelity [13]. VAEs decompose the compression problem into two components: (1) a probabilistic encoder that maps input embeddings to a latent distribution $q_\phi(\mathbf{z} \mid \mathbf{x})$, and (2) a decoder f_{dec} that reconstructs embeddings from samples drawn from this distribution.

The VAE training objective minimizes the negative evidence lower bound (negative ELBO), combining a reconstruction term with a Kullback–Leibler (KL) divergence regularizer:

$$\mathcal{L}_{\text{VAE}} = -\,\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\big[\log p_\theta(\mathbf{x} \mid \mathbf{z})\big] + \beta\,\mathrm{KL}\big(q_\phi(\mathbf{z} \mid \mathbf{x}) \,\|\, p(\mathbf{z})\big)\,, \tag{1}$$

where β controls the trade-off between reconstruction fidelity and latent-space regularization [13]. Higher β values encourage more compact representations at the risk of degraded reconstruction quality.

Application to Audio Embeddings: VAEs and related adversarial autoencoders can effectively compress high-dimensional acoustic features [13], [21]. However, because reconstruction operates in the original embedding space, the code optimizes for general-purpose fidelity rather than task-relevant class separability.

2) SUPERVISED BOTTLENECK AUTOENCODERS

To address this limitation, **supervised bottleneck autoencoders** jointly optimize reconstruction and classification objectives [22]. The architecture consists of an encoder $f_{\text{enc}}(\mathbf{x}) = \mathbf{z}$, a classifier head $g_{\text{class}}(\mathbf{z}) = \hat{\mathbf{y}}$, and a decoder $f_{\text{dec}}(\mathbf{z}) = \hat{\mathbf{x}}$.

The combined objective is:

$$\mathcal{L}_{\text{bottleneck}} = \mathcal{L}_{\text{recon}}(\mathbf{x}, \hat{\mathbf{x}}) + \lambda_{\text{clf}}\,\mathcal{L}_{\text{CE}}(\mathbf{y}, \hat{\mathbf{y}}), \tag{2}$$

where λ_{clf} weights classification loss relative to reconstruction loss [22]. By enforcing that the bottleneck $\mathbf{z}$ supports both reconstruction and accurate classification, the model preserves task-discriminative features while eliminating semantic redundancy.

Theoretical and Empirical Support: Le et al. [22] formally prove a uniform-stability generalization bound for supervised autoencoders, establishing that the reconstruction objective acts as a data-dependent regularizer that provably reduces the generalization gap. They further demonstrate empirically that adding reconstruction loss to a supervised

network never degrades and frequently improves classification performance. This theoretical guarantee is directly applicable for our setting, where only 654 training clips are available: the reconstruction term provides principled regularization beyond standard dropout, ensuring that the task-discriminative bottleneck generalizes reliably from limited data.

C. QUANTIZATION TECHNIQUES FOR DISCRETE REPRESENTATIONS

1) PRODUCT QUANTIZATION

Product Quantization (PQ) partitions high-dimensional vectors into subvectors and quantizes each independently [11]. For a D-dimensional embedding divided into M subvectors of dimension $d = D/M$, the procedure trains a k-means codebook $\mathbf{C}_m = \{\mathbf{c}_{m,1}, \ldots, \mathbf{c}_{m,K}\}$ with $K = 256$ codewords per subgroup and replaces each subvector with its nearest-codeword index ($\log_2 K$ bits). The quantized representation uses $M \log_2 K$ bits total [11]; for example, a 256-D embedding with $M = 8$ and $K = 256$ requires $8 \times 8 = 64$ bits, a 128× reduction relative to its 1024-byte FP32 form.

Limitations: PQ is inherently unsupervised—codewords optimize reconstruction quality, not class separability. At high compression ratios, the independence assumption (quantizing subvectors separately) ignores inter-dimension correlations, which can cause irreversible information loss for fine-grained classification [12].

2) OPTIMIZED PRODUCT QUANTIZATION

Optimized Product Quantization (OPQ) improves upon PQ by learning an orthogonal rotation matrix $\mathbf{R}$ that redistributes the variance of embedding dimensions before quantization, thereby minimizing the overall reconstruction error [12]:

$$\begin{aligned} \min_{\mathbf{R},\, \mathbf{C}_1,\ldots,\mathbf{C}_M} & \sum_{i=1}^{N} \|\mathbf{R}\mathbf{x}_i - \mathbf{c}_i\|^2 \\ \text{s.t.} \quad & \mathbf{R}^T\mathbf{R} = \mathbf{I}, \\ & \mathbf{c}_i \in \mathbf{C}_1 \times \cdots \times \mathbf{C}_M \end{aligned} \tag{3}$$

where N is the number of training vectors, $\mathbf{x}_i$ denotes the ith input embedding, $\mathbf{R}$ is the learned orthogonal rotation matrix, $\mathbf{C}_1, \ldots, \mathbf{C}_M$ are the subspace codebooks, and $\mathbf{c}_i$ is the quantized representation of $\mathbf{x}_i$ obtained by selecting one centroid from each subspace codebook. The orthogonality constraint $\mathbf{R}^T\mathbf{R} = \mathbf{I}$ preserves Euclidean distances after rotation. The reconstructed vector is given by $\hat{\mathbf{x}}_i(\mathbf{R}) = \text{PQ}(\mathbf{R}\mathbf{x}_i)$. The rotation matrix and codebooks are optimized alternately, yielding approximately 10–20% lower reconstruction error than conventional PQ on standard retrieval benchmarks [12].

3) VECTOR QUANTIZED VARIATIONAL AUTOENCODERS

VQ-VAE combines VAEs with discrete quantization [23]. A learnable codebook $\mathbf{E} \in \mathbb{R}^{K \times D_e}$ stores K codewords. The encoder output is assigned to the nearest codebook entry via:

$$k^* = \arg\min_k \|\mathbf{z}_e - \mathbf{e}_k\|^2, \quad \mathbf{e} \leftarrow \mathbf{e}_{k^*}, \tag{4}$$

where k^* is the winning index and $\mathbf{e}_{k^*}$ is the corresponding codebook vector. The VQ-VAE loss, where sg[·] denotes the stop gradient operator (i.e., the identity in the forward pass with zero gradient in the backward pass), is:

$$\mathcal{L}_{\text{VQ-VAE}} = \|\mathbf{x} - \hat{\mathbf{x}}\|^2 + \|\,\text{sg}[\mathbf{z}_e] - \mathbf{e}\|^2 + \beta\|\mathbf{z}_e - \text{sg}[\mathbf{e}]\|^2, \tag{5}$$

where the second (codebook) term moves the selected codeword toward the encoder output, and the third (commitment) term, weighted by β, prevents encoder drift [23]. VQ-VAE enables end-to-end learnable compression with interpretable discrete codes, a key advantage over fixed quantization methods.

D. TRANSFER LEARNING AND JOINT OPTIMIZATION STRATEGIES

1) FINE-TUNING PARADIGMS FOR DOMAIN ADAPTATION

Fine-tuning pre-trained audio models on marine mammal data has become standard practice [24]. The typical approach involves initializing from AudioSet- or YouTube-8M-pretrained weights, attaching a task-specific head (e.g., 15-class softmax), and updating all or a subset of weights on the target dataset [24]. Full fine-tuning enables greater representational shift but risks overfitting when target data is limited. Recent empirical studies on self-supervised transformer training suggest that careful learning-rate scheduling and hyperparameter tuning enable effective full fine-tuning even with modest target datasets [25].

2) END-TO-END JOINT TRAINING WITH TASK-AWARE LOSSES

Rather than treating embedding extraction, compression, and classification as separate stages, **end-to-end joint training** optimizes all components jointly [26]. The unified loss couples reconstruction and classification:

$$\mathcal{L}_{\text{joint_lit}} = \mathcal{L}_{\text{class}}(\mathbf{y}, \hat{\mathbf{y}}) + \alpha\, \mathcal{L}_{\text{recon}}(\mathbf{x}, \hat{\mathbf{x}}) + \beta\, \mathcal{L}_{\text{KL}}(q(\mathbf{z}), p(\mathbf{z})), \tag{6}$$

where α and β control reconstruction and regularization trade-offs [26]. Information Bottleneck analysis [18] predicts that such training reduces mutual information between the compressed code and the input while preserving mutual information with the target label.

a: EMPIRICAL EVIDENCE

Jointly trained autoencoders have achieved substantial compression with minimal accuracy loss on vision benchmarks [26], and deep convolutional networks with data augmentation achieve strong environmental-sound classification accuracy [27]—demonstrating that task-coupled objectives

preserve discriminative accuracy under aggressive model compression.

b: DISTINCTION FROM THE PROPOSED FRAMEWORK

The joint-training strategies above [26], [28] optimize encoder–classifier pipelines end-to-end from raw inputs, discarding the priors of large pretrained models. Our work differs in three ways. First, it operates on *frozen* foundation-model embeddings—the representational capacity of Perch is preserved while a lightweight bottleneck distills a task-specific subspace. Second, the objective is driven by a hard byte-budget imposed by the acoustic channel (8 bytes at 32 bps), rather than a generic compression ratio. Third, the compression stage is co-designed with transmission-aware quantization (PQ/OPQ) so that the final payload matches commercial modem frame sizes. To our knowledge, no prior joint-training pipeline has addressed this combination of frozen-foundation-model distillation driven by an underwater-telemetry byte budget.

E. SEMANTIC REDUNDANCY AND THE GENERALIZATION TAX

The **generalization tax** [8] describes how broadly trained models learn general-purpose representations to perform well across diverse tasks, at the cost of efficiency on any single narrow application. For domain-specific tasks like marine mammal classification, much of the representational capacity is therefore wasted on task-irrelevant features [29], dominated by redundant ''recording-specific characteristics'' (i.e., background noise and environmental artifacts) [29]. While audio foundation models optimized for broad objectives such as AudioSet tagging [30] achieve strong cross-domain performance, Hummel et al. [29] demonstrate directly that pretrained audio embeddings applied to underwater acoustic target recognition contain substantial task-irrelevant structure that degrades narrow-task class separability—precisely the redundancy our framework eliminates. This phenomenon is termed **semantic redundancy**: high-dimensional embeddings contain information about task-irrelevant sound attributes (e.g., background noise characteristics, recording device artifacts) that inflate the representation size without aiding classification.

Task-Aware Compression as Redundancy Elimination: By jointly optimizing compression and classification, task-aware pipelines explicitly eliminate semantic redundancy. An embedding dimension encoding subtle variations in ambient sound (irrelevant for species identification) would be discarded in a joint-training regime, whereas unsupervised compression (PQ, VAE) would attempt to preserve it. Empirically, this explains why supervised bottleneck autoencoders achieve higher compression-accuracy trade-offs than unsupervised alternatives [22], [26].

F. GAPS AND RESEARCH OPPORTUNITIES

Several critical gaps remain in the literature:

1) **Limited marine-specific benchmarks:** Most foundation model research (Perch, PaSST, PANNs) focuses on avian or general audio. Systematic evaluation on marine mammal datasets is sparse, making model selection uncertain.
2) **Compression–classification trade-off characterization:** Individual compression techniques and fine-tuning approaches are well-studied, but their interactions in an end-to-end pipeline remain poorly understood.
3) **Verifiability:** Compressed representations at extreme ratios may suffer from irreversible information loss. Systematic ablation studies are scarce.
4) **Telemetry constraints:** No prior work explicitly co-designs a compression pipeline with the bandwidth limitations of acoustic modems (64–640 bps).
5) **Temporal dynamics:** Most compression techniques treat audio as static clips. Frame-level representations (as in Perch) may unlock better compression-accuracy trade-offs but remain underexplored for marine applications.

III. METHODOLOGY

This section details the proposed task-aware compression pipeline. The design is governed by two primary constraints of the marine PAM context: a 654-clip training corpus and a hard 8-byte transmission budget imposed by acoustic modem bandwidth limits. The following subsections outline the pipeline sequentially: dataset curation, embedding extraction, supervised bottleneck autoencoder, classification on compressed representations, quantization, and evaluation metrics.

A. DATASET CURATION AND PREPROCESSING

We curated a 15-class subset from the WMMD [14], balancing statistical robustness with acoustic-telemetry constraints, as shown in Table 3. The selection was driven by three exclusion criteria.

1) SAMPLE SIZE SUFFICIENCY

We imposed a minimum of $N \geq 28$ clips per class for reliable stratification. Species with insufficient samples—such as the *Weddell Seal* ($N = 2$), *Leopard Seal* ($N = 8$), and *Minke Whale* ($N = 14$)—were excluded. The final subset ranges from 28 clips (*Killer Whale*) to 60 clips (*Sperm Whale*).

2) SPECTRAL COMPATIBILITY WITH TELEMETRY CONSTRAINTS

All audio was standardized to 32 kHz (16 kHz Nyquist). Exploratory Data Analysis confirmed that the median 95th-percentile spectral energy across the subset is 13.5 kHz, well within the Nyquist limit, preserving the discriminative features required for classification.

3) PHYLOGENETIC AND VOCAL DIVERSITY

The 15 classes span three taxonomic groups to test the foundation models against a wide range of call types:

- **Mysticetes (Baleen Whales):** Low-frequency moans (e.g., *Fin Whale*, 90% of energy < 500 Hz).
- **Odontocetes (Toothed Whales):** Mid-to-high frequency clicks and whistles (e.g., *False Killer Whale*, 59% of energy in 5–16 kHz).
- **Pinnipeds:** Distinct amphibious vocal structures (e.g., *Walrus*, *Bearded Seal*).

TABLE 3. The 15-class marine mammal subset. Train *N* values sum to 654 (the held-out test split adds 165 clips, for 819 total). Frequency bands are defined by 95% spectral energy concentration.

Species	Order	Train N	Freq. Band
Sperm Whale	Odontocete	60	Mid-High
Long-Finned Pilot Whale	Odontocete	56	Mid
Short-Finned Pilot Whale	Odontocete	53	Mid
Humpback Whale	Mysticete	51	Low
Melon-Headed Whale	Odontocete	50	High
Bowhead Whale	Mysticete	48	Low
False Killer Whale	Odontocete	47	High
Northern Right Whale	Mysticete	43	Low
Ross Seal	Pinniped	40	Mid
Beluga Whale	Odontocete	40	Low-Mid
Fin Whale	Mysticete	40	Low
Harp Seal	Pinniped	38	Low-Mid
Bearded Seal	Pinniped	30	Low-Mid
Walrus	Pinniped	30	Low-Mid
Killer Whale	Odontocete	28	Mid

All audio is resampled to 32 kHz and normalized to 2-second clips (64,000 samples). Audio is mean-centered and peak-normalized to $[-1, 1]$. We split data into 654 training clips and 165 test clips, with stratification by species and recording session to avoid leakage.

B. EMBEDDING EXTRACTION AND EVALUATION

We compare three embedding extractors, all frozen (inference mode only):

1) VGGISH

A shallow CNN trained on YouTube-8M (2M+ video-audio pairs), producing 128-dimensional vectors from log-Mel spectral features. Despite computational efficiency, VGGish exhibits poor separation on marine mammal sounds due to domain mismatch.

2) PASST

A transformer-based encoder outputting 768-dimensional embeddings via multi-head self-attention on log-Mel spectrogram patches [10]. Demonstrates superior performance on audio classification benchmarks compared to VGGish, though still trained without explicit marine content.

3) PERCH (GOOGLE'S BIOACOUSTIC FOUNDATION MODEL)

A 12M-parameter EfficientNet-based model pre-trained on a large, curated corpus of bird and wildlife recordings spanning thousands of species [7]. It outputs 1536-dimensional embeddings capturing both coarse and fine-grained acoustic features. Despite minimal explicit marine mammal data in pre-training, Perch's broad bioacoustic coverage enables strong transfer to the marine domain, as our baseline comparison confirms.

All embeddings are extracted at the clip level and standardized using scikit-learn's `StandardScaler` fitted on training embeddings.

C. SUPERVISED BOTTLENECK AUTOENCODER

To compress embeddings while preserving task-discriminative information, we train a supervised bottleneck autoencoder with three components:

Encoder: Linear $1536 \to 512$, ReLU, dropout (0.1), Linear $512 \to d$, where $d \in \{64, 128, 256, 512\}$.

Decoder: Linear $d \to 512$, ReLU, dropout (0.1), Linear $512 \to 1536$.

Classification head: Linear $d \to 256$, ReLU, dropout (0.2), Linear $256 \to 15$ (softmax).

The base joint loss function is:

$$\mathcal{L}_{\text{joint}} = \lambda_{\text{recon}}\,\mathcal{L}_{\text{MSE}}(\hat{\mathbf{e}}, \mathbf{e}) + \lambda_{\text{clf}}\,\mathcal{L}_{\text{CE}}(\hat{\mathbf{y}}, y), \tag{7}$$

where $\mathbf{e}$ is the original Perch embedding, $\hat{\mathbf{e}}$ is the reconstruction, y is the species label, and $\lambda_{\text{recon}}, \lambda_{\text{clf}}$ control the reconstruction–classification trade-off. For the multi-task ablation (Section IV-F), we additionally include a cosine-similarity regularization term:

$$\mathcal{L}_{\text{joint3}} = \lambda_{\text{recon}}\,\mathcal{L}_{\text{MSE}} + \lambda_{\text{clf}}\,\mathcal{L}_{\text{CE}} + \lambda_{\cos}\big(1 - \cos(\mathbf{e}, \hat{\mathbf{e}})\big), \tag{8}$$

evaluated under three weighting regimes: *Recon-Heavy* $(\lambda_{\text{recon}}, \lambda_{\text{clf}}, \lambda_{\cos}) = (1.0, 0.3, 0.5)$, *Balanced* $= (0.5, 0.5, 0.3)$, and *Class-Focused* $= (0.2, 0.8, 0.1)$.

For the two-term ablation (Section IV-E), we swept $\lambda_{\text{recon}} \in \{0.5, 0.7, 1.0\}$ and $\lambda_{\text{clf}} \in \{0.0, 0.1, 0.3, 0.5, 0.7, 1.0\}$ via cross-validation, selecting $\lambda_{\text{recon}} = 1.0$, $\lambda_{\text{clf}} = 0.5$ as the optimal balance.

The model is trained for 20 epochs using Adam (lr $= 10^{-3}$, weight decay $= 10^{-5}$), batch size 256, with early stopping (patience $= 5$) on validation loss from a 90/10 train/val split.

Computational and energy overhead: The bottleneck encoder adds negligible edge cost relative to Perch. Its two linear layers ($1536 \to 512 \to d$) require approximately 0.9–1.0 million multiply-accumulate operations (MACs) per 2-second clip, which is less than 0.2% of the ≈500M MACs required by the Perch EfficientNet inference. The subsequent OPQ codebook lookup adds a further ≈65,000 MACs. In a deployed PAM sensor, the hydrophone array, signal conditioning amplifier, ADC, and Perch inference dominate energy consumption, whereas the bottleneck AE contributes only a negligible fraction. The AE encoder therefore imposes no meaningful additional constraint on the battery budget of a long-duration moored deployment. **Dataset scale and justification for a deep learning approach:** Three factors address potential overfitting concerns on 654 training clips. First, the foundation model (Perch) is *completely frozen*:

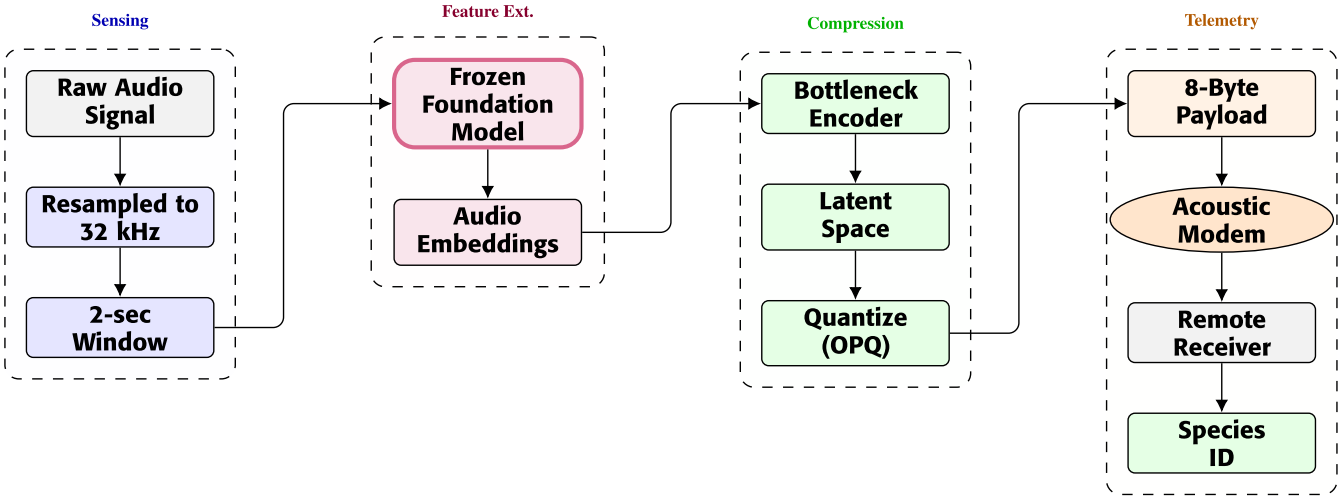


FIGURE 1. Linear underwater acoustic signal-processing flow: sensing, frozen foundation-model feature extraction, task-aware compression (bottleneck + OPQ), and telemetry to a remote receiver.

only the AE's encoder–decoder (≈1.6M parameters for $d = 256$) is trained—a parameter count comparable in practice to a kernel SVM on the 1536-D Perch features. Second, the reconstruction loss acts as a data-dependent regularizer (as formally proven by Le et al. [22]), helping to prevent latent-space collapse and improve generalization. Consistent with this effect, several configurations in Table 9 achieve retention rates exceeding 100%, suggesting that the compressed latent representations are not suffering from substantial overfitting. Third, Table 5 benchmarks traditional linear compression (PCA): PCA-128D attains 98.2% retention at 12× compression, but falls sharply to 93.2% at 24×, confirming that linear projections are insufficient for the byte budgets required by acoustic modems. The supervised AE, operating on the same frozen Perch features, achieves 97.1% retention at 768×—demonstrating the necessity of non-linear, task-aware compression beyond what PCA or direct scalar quantization can provide.

D. CLASSIFICATION ON COMPRESSED REPRESENTATIONS

We evaluate classification in two modes. In **latent classification**, the classifier head operates directly on $\mathbf{z} = f_{\text{enc}}(\mathbf{e})$ at test time without reconstruction, isolating compression-induced loss. In **reconstructed classification**, latent vectors are decoded to $\hat{\mathbf{e}}$ and classified by a separate logistic regression model, establishing whether unsupervised classifiers can adapt to the reconstructed space.

E. QUANTIZATION METHODS

After learning compact latent representations, we apply vector quantization to further reduce transmission costs. For **PQ** we partition the latent into m subvectors and independently quantize each with k-means ($k = 256$); the code uses $8m$ bits. For a 256-D latent with $m = 8$:

$$B_{\text{PQ}} = 8 \times 8 \text{ bits} = 64 \text{ bits} = 8 \text{ bytes}. \tag{9}$$

The choice of $m = 8$ subvectors ensures the compressed code fits a strict payload budget imposed by underwater telemetry. Each 2-second clip yields 8 subvectors × 8 bits = 64 bits = 8 bytes. When transmitted at the clip rate (one clip per 2 seconds), this payload translates to a transmission bitrate of 64 bits/2 seconds = 32 bps which falls well below the 64-bps reliable floor of commercial acoustic modems (Table 1), leaving headroom for metadata. For **OPQ**, we minimize quantization distortion via an alternating orthogonal-rotation optimization over 10 iterations:

$$\min_{\mathbf{R}} \sum_{i=1}^{N} \|\mathbf{z}_i - \text{PQ}(\mathbf{R}\mathbf{z}_i)\|^2, \tag{10}$$

fixing $\mathbf{R}$ to update codebooks via k-means, then fixing the codebooks to update $\mathbf{R}$ via the Orthogonal Procrustes problem.

F. EVALUATION METRICS

We report weighted F1 (support-weighted average across classes); F1 retention (percentage of the FP32 baseline F1 preserved); **compression ratio** $\text{CR} = (1536 \times 32 \text{ bits})/B_{\text{code}}$, referenced to the 6144-byte FP32 embedding; and **transmission bitrate** $R = B_{\text{code}}/T$ where $T = 2$ s, enabling direct comparison with acoustic-modem bandwidth (64–640 bps).

IV. RESULTS

A. EMBEDDING MODEL COMPARISON

We first evaluate the three embedding families without compression to establish full-precision baselines. As shown in Table 4, Perch (1536-D) with Logistic Regression achieves the highest performance at 0.9149 weighted F1. PaSST (768-D) reaches 0.8500, and VGGish (128-D) requires LightGBM to achieve 0.8465. The +0.0649 gap between Perch and PaSST indicates that Perch's bioacoustics-focused pre-training yields a far more linearly separable latent space

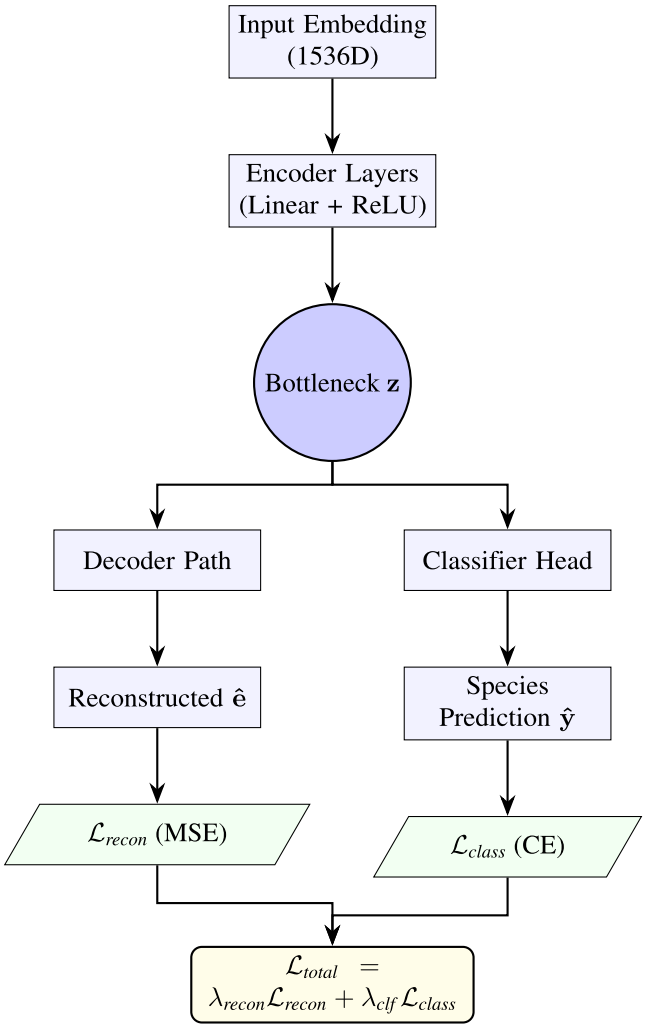


FIGURE 2. Supervised bottleneck autoencoder architecture. The shared bottleneck z feeds both a reconstruction decoder and a classification head, trained jointly via Eq. (7).

TABLE 4. Baseline classifier performance on full-precision embeddings. Perch (1536-D) provides the highest F1 and supports linear separability, making it the optimal foundation for compression.

Embedding Model	Dim.	Best Classifier	Test F1
VGGish	128	LightGBM	0.8465
PaSST	768	Logistic Regression	0.8500
Perch	1536	Logistic Regression	0.9149

for this marine domain. We therefore select Perch for all downstream compression experiments.

B. COMPRESSION BASELINE: DIRECT QUANTIZATION

Before introducing neural compression, we establish baselines from direct dimensionality reduction and quantization on full-precision Perch embeddings. INT8 quantization (4×) is "free"—100.00% retention. INT4 (8×) drops negligibly to 99.37%. PCA maintains over 98% retention up to 12× but falls sharply at 24× (93.15%), indicating linear projections discard critical variance beyond a bottleneck. PQ extends this boundary, achieving 97.78% retention at 384× but 95.02% at 768×. These baselines show that direct quantization reaches high compression ratios with diminishing fidelity; the learned non-linear autoencoder reaches the same byte budgets at higher retention while additionally discarding task-irrelevant structure.

TABLE 5. Direct quantization baselines on Perch embeddings. All compression ratios use a unified FP32 reference (1536-d × 4 bytes = 6144 bytes per vector).

Method	Comp.	F1 Score	F1 Ret.
Baseline (FP32)	1×	0.9149	100.00%
INT8 Quantization	4×	0.9149	100.00%
INT4 Quantization	8×	0.9091	99.37%
PCA-128D	12×	0.8984	98.20%
PCA-64D	24×	0.8522	93.15%
PQ-16×256	384×	0.8945	97.78%
PQ-8×256	768×	0.8693	95.02%

C. SUPERVISED VS. UNSUPERVISED AUTOENCODER

Table 6 compares autoencoders trained with pure reconstruction versus joint classification across 7 independent seeds. Task-aware training yields consistent, statistically robust improvements: at the 128-D bottleneck (12×), the supervised AE outperforms the unsupervised counterpart by +2.91% F1 retention (95.32 ± 1.04% vs. 92.41%), with the separation confirmed across all 7 seeds. At 256-D (6×), the gain is +1.16% (94.62 ± 0.90%). The 512-D gap narrows to +0.12%, consistent with wider bottlenecks having less capacity to waste on noise. This validates our core hypothesis: an unsupervised AE wastes latent capacity reconstructing task-irrelevant environmental noise (e.g., ocean currents, vessel noise), whereas joint optimization selectively discards it while preserving the harmonic features required to separate marine mammal classes. The 256-D supervised AE provides the most stable foundation for downstream quantization pipelines.

D. PRODUCT QUANTIZATION VS. OPTIMIZED PRODUCT QUANTIZATION

Table 7 compares PQ and OPQ applied to the supervised-AE latents. The results reveal a geometry-dependent pattern: at the 384× operating point (AE-512D + 16 subvectors), OPQ delivers a substantial +2.30% retention gain, reaching 99.32% (F1 0.9087)—the highest accuracy in any quantized configuration. At the headline 768× point (AE-256D + 8 subvectors), the OPQ gain is smaller (+0.07%, reaching 97.14%) as the 256-D latent is already relatively well-balanced. The one exception is AE-384D at 512×, where PQ (98.57%) marginally outperforms OPQ (98.37%), consistent with this intermediate dimension having naturally decorrelated subvectors. Overall, OPQ is the preferred

TABLE 6. Supervised vs. unsupervised autoencoder performance (mean ± std, $N = 7$ seeds). Task-aware training consistently outperforms pure reconstruction at tight bottlenecks; the confidence intervals confirm the gain at 128-D and 256-D is robust across runs.

Bottleneck	Comp.	Unsupervised F1	Supervised F1	Sup. Ret. (±std)
128D	12.0×	0.8455 ± 0.0084	0.8721 ± 0.0095	95.32 ± 1.04%
256D	6.0×	0.8551 ± 0.0069	0.8657 ± 0.0083	94.62 ± 0.90%
512D	3.0×	0.8704 ± 0.0088	0.8715 ± 0.0118	95.25 ± 1.29%

deployment choice: the orthogonal rotation adds negligible inference overhead while providing consistent or superior retention.

TABLE 7. PQ vs. OPQ on supervised-AE latents. OPQ gains are geometry-dependent: large at 384×, negligible at 768×. All F1 scores are 3-run means.

Configuration	Comp.	PQ F1	PQ Ret.	OPQ F1	OPQ Ret.	OPQ Gain
AE-512D + 16 subvec.	384×	0.8877	97.02%	0.9087	99.32%	+2.30%
AE-384D + 12 subvec.	512×	0.9018	98.57%	0.9000	98.37%	−0.20%
AE-256D + 8 subvec.	768×	0.8882	97.08%	0.8888	97.14%	+0.07%

E. TASK LOSS WEIGHT ABLATION

Table 8 reports the effect of sweeping λ_{clf} over {0.0, 0.1, 0.3, 0.5, 0.7, 1.0} at fixed $\lambda_{\text{recon}} = 1.0$ for the 256-D bottleneck. The results confirm the inverted-U pattern expected from IB theory: purely unsupervised ($\lambda_{\text{clf}} = 0.0$) yields only 92.58% retention; introducing classification pressure improves retention monotonically up to $\lambda_{\text{clf}} = 0.3$ (96.69%), after which excessive classification pressure begins to suppress the reconstruction regularizer. At $\lambda_{\text{clf}} = 0.7$, retention drops to 92.84%, recovering only if the classification loss completely dominates at $\lambda_{\text{clf}} = 1.0$ (95.94%), where the model reduces to a pure discriminative bottleneck. Note that with only two runs at each λ value, the high variance at $\lambda = 0.3$ ($\sigma = 0.0099$) means the ordering between $\lambda = 0.3$ and $\lambda = 0.5$ should be interpreted cautiously; both values outperform all alternatives by a clear margin. Our downstream experiments use $\lambda_{\text{clf}} = 0.5$, which provides a robust, low-variance operating point within the optimal regime.

F. END-TO-END JOINT TRAINING WITH MULTI-TASK LOSSES

Tab. 9 reports the three-term loss of Eq. (8) across bottleneck dimensions and weighting regimes. Several configurations reach or exceed the full-precision baseline—the *Recon-Heavy* setup at 256-D reaches 100.61% and the *Balanced* setup at 128-D reaches 100.07%.

Information Bottleneck Principle: Consider the Markov chain $\mathbf{x} \to \mathbf{z} \to \mathbf{y}$, where:

- $\mathbf{x}$ is the original high-dimensional embedding,
- y is the target species label (class), and
- $\mathbf{z}$ is the compressed bottleneck representation.

The Information Bottleneck (IB) principle [18] characterizes an optimal compression as one that maximizes mutual information with the target label $I(\mathbf{z}; y)$ while minimizing information retention from the input $I(\mathbf{z}; \mathbf{x})$. Intuitively, $\mathbf{z}$ should capture only the features necessary for predicting y, discarding task-irrelevant aspects of $\mathbf{x}$ (such as recording artifacts). This is formalized by the IB trade-off curve: for a fixed bandwidth on $\mathbf{z}$, maximizing $I(\mathbf{z}; y)$ requires minimizing $I(\mathbf{z}; \mathbf{x})$. The IB principle [18] characterizes an ideal latent code as one that maximizes $I(\mathbf{z}; y)$ (mutual information with the label) while minimizing $I(\mathbf{z}; \mathbf{x})$ (retained information from the input). Fig. 3 plots estimated $I(\mathbf{z}; \mathbf{x})$ vs. $I(\mathbf{z}; y)$ across 20 training epochs for $d \in \{128, 256, 512\}$, computed via binning-based MI estimation on the training set. All three curves exhibit the two-phase dynamics identified by Shwartz-Ziv and Tishby [31]: a rapid *fitting* phase (epochs 1–6) during which $I(\mathbf{z}; y)$ rises sharply to a peak of 0.92–0.98, followed by a *compression* phase (epochs 6–20) during which $I(\mathbf{z}; \mathbf{x})$ decreases by 0.012–0.014 bits as the encoder discards task-irrelevant input dimensions. The 128-D bottleneck achieves the highest $I(\mathbf{z}; y)$ peak (0.9775) but compresses most aggressively, consistent with its tighter capacity constraint forcing stronger input-dimension selection. All dimensions converge to low $I(\mathbf{z}; \mathbf{x})$ (≈0.11 bits) while sustaining substantial $I(\mathbf{z}; y)$ (≈0.75–0.82 bits), confirming that the compression is semantically principled: recording-condition variance (vessel noise, ambient reverberation) is progressively discarded while species-discriminative harmonic structure is retained. This directly explains the >100% retention results above: the baseline Logistic Regression fitted on full-dimensional

TABLE 8. Ablation of λ_{clf} (256-D bottleneck, $\lambda_{\text{recon}} = 1.0$, 2-run mean). The inverted-U curve confirms that balancing reconstruction and classification is critical; $\lambda \in \{0.3, 0.5\}$ both lie within the optimal regime.

λ_{clf}	F1 (mean)	F1 (std)	F1 Retention
0.0	0.8470	0.0030	92.58%
0.1	0.8629	0.0003	94.32%
0.3	**0.8846**	0.0099	**96.69%**
0.5	0.8731	0.0005	95.44%
0.7	0.8494	0.0001	92.84%
1.0	0.8778	0.0033	95.94%

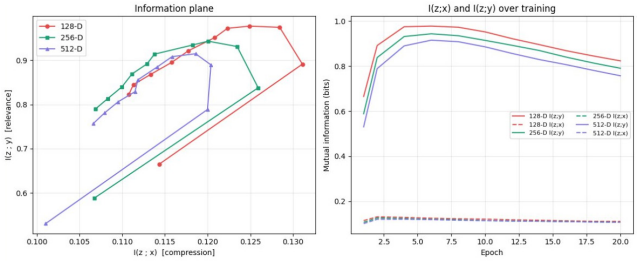


FIGURE 3. Information-plane analysis of the supervised bottleneck AE for $d \in \{128, 256, 512\}$. Left: $I(\mathbf{z}; \mathbf{x})$–$I(\mathbf{z}; y)$ trajectories across 20 epochs; all curves exhibit the IB fitting-then-compression two-phase pattern (an initial rise in $I(\mathbf{z}; y)$ followed by a sustained decrease in $I(\mathbf{z}; \mathbf{x})$). Right: epoch-by-epoch $I(\mathbf{z}; y)$ (solid) and $I(\mathbf{z}; \mathbf{x})$ (dashed); $I(\mathbf{z}; \mathbf{x})$ decreases monotonically after epoch 6, confirming progressive input compression while label relevance is maintained.

Perch embeddings was overfitting to recording artifacts; the IB-guided bottleneck removes those artifacts from the code, yielding a downstream classifier that generalizes better than the uncompressed baseline—an outcome consistent with the supervised-autoencoder regularization analysis of Le et al. [22].

In bioacoustic monitoring, species-discriminative features (harmonic structure, pulse repetition rate, frequency contour) occupy a small subspace of total acoustic information, while recording artifacts (vessel noise, depth-dependent absorption, hydrophone self-noise) occupy the complementary subspace. The measured decrease in $I(\mathbf{z}; \mathbf{x})$ during the compression phase directly quantifies the elimination of this recording-condition variance, confirming that our compression is semantically driven rather than arbitrary.

TABLE 9. End-to-end joint training (Eq. 8). Regimes: Recon-Heavy ($\lambda_{recon}, \lambda_{clf}, \lambda_{cos}$) = (1.0, 0.3, 0.5); Balanced = (0.5, 0.5, 0.3); Class-Focused = (0.2, 0.8, 0.1). Retention ≈ 100% reflects regularization, not information gain.

Regime	Bottleneck	Comp.	F1	Retention
Recon-Heavy	128D	12.0×	0.8829	96.50%
	256D	6.0×	0.9204	**100.61%**
	512D	3.0×	0.8953	97.86%
Balanced	128D	12.0×	0.9155	**100.07%**
	256D	6.0×	0.8957	97.90%
	512D	3.0×	0.8830	96.51%
Class-Focused	128D	12.0×	0.8956	97.89%
	256D	6.0×	0.8999	98.36%
	512D	3.0×	0.8942	97.74%

G. EXTREME COMPRESSION AND MODEM COMPATIBILITY

Table 10 presents the complete pipeline from the raw Perch embedding down to the final modem-compatible payload. The 256-D supervised AE alone (1024 bytes, 6×) achieves 96.40% retention. Adding OPQ to reach 8 bytes (768×) yields 97.14% retention (F1 0.8888)—our headline deployment configuration. At a 2-second clip cadence, this 64-bit payload corresponds to just 32 bps, comfortably within the operating range of entry-level acoustic modems (Table 1).

At 4 bytes (AE-256D + PQ-4 × 128, 1536×, 16 bps), using a 128-codeword codebook (7 bits per subvector), the pipeline achieves 94.30% retention (F1 0.8627). Both 8-byte and 4-byte payloads are below the floor of the Water Linked M64 modem (64 bps), leaving headroom for metadata (GPS timestamp, depth, hydrophone gain) at no additional bandwidth cost.

H. SUMMARY OF RESULTS

Five findings emerge from our evaluation:

1) **Perch provides the strongest foundation:** 0.9149 F1 with a linear probe, far outperforming PaSST (0.8500) and VGGish (0.8465) on the marine domain.

TABLE 10. Extreme-compression pipeline stages targeting underwater modem bandwidths. The bold row is the recommended deployment point. The 4-byte PQ configuration achieves 94.30% retention at 16 bps using a 128-codeword codebook, extending coverage to sub-32 bps links.

Configuration	Payload	Comp.	Bitrate	F1	Retention
Baseline (Perch FP32)	6144 B	1×	—	0.9149	100.00%
INT8 Scalar Quantization	1536 B	4×	—	0.9149	100.00%
Supervised AE-256D (no quant.)	1024 B	6×	—	0.8820	96.40%
Sup-AE-512D + PQ-16×256	16 B	384×	64 bps	0.8877	97.02%
Sup-AE-384D + PQ-12×256	12 B	512×	48 bps	0.9018	98.57%
Sup-AE-256D + OPQ-8×256	**8 B**	**768×**	**32 bps**	**0.8888**	**97.14%**
Sup-AE-256D + PQ-4×128	4 B	1536×	16 bps	0.8627	94.30%

2) **Task-aware compression is strictly necessary:** Supervised autoencoders outperform unsupervised reconstruction by +0.12% to +2.91% F1 retention across bottleneck dimensions (7-seed mean, Table 6), confirming that joint optimization effectively discards environmental noise.
3) **Balanced multi-task training adheres to the IB principle:** The ablation identifies $\lambda_{clf} \in \{0.3, 0.5\}$ as the optimal regime; quantitative IB analysis (Fig. 3) shows the fitting-then-compression two-phase dynamics predicted by Tishby [18], and the >100% retention results in multi-task training are explained by removal of overfitting-inducing recording artifacts from the latent code.
4) **OPQ's benefit is geometry-dependent:** Gains range from +2.30% at 384× to negligible at 768×; in all cases OPQ is preferred given its negligible inference overhead.
5) **Both 8-byte and 4-byte payloads are viable for acoustic modems:** The 8-byte, 768× OPQ configuration achieves 97.14% F1 retention at 32 bps; the 4-byte, 1536× PQ configuration (AE-256D + PQ-4 × 128) achieves 94.30% retention at 16 bps, both well within the 64 bps floor of commercial acoustic modems.

V. CONCLUSION

We present a task-aware compression framework that reconciles the accuracy of audio foundation models with the severe bandwidth constraints of underwater acoustic telemetry. By distilling frozen Perch embeddings through a supervised bottleneck autoencoder and quantizing the result with Optimized Product Quantization, we reduce a 6144-byte FP32 embedding to just 8 bytes per 2-second clip—a 768× reduction—while retaining 97.14% of the baseline F1. At the most aggressive extreme, a 4-byte PQ configuration (AE-256D + PQ-4 × 128) achieves 94.30% retention at 1536× compression and 16 bps. Crucially, quantitative information-plane analysis confirms that training follows the IB fitting-then-compression trajectory: $I(\mathbf{z}; \mathbf{x})$ decreases systematically after epoch 6 as recording artifacts are stripped from the code, while $I(\mathbf{z}; y)$ remains high—empirically confirming that the discarded information is redundant for the target task and that the "generalization tax"

of broadly-trained foundation models can be substantially refunded for domain-specific deployment. Unlike on-edge recognition, the transmitted representation remains verifiable, re-scorable, and re-trainable server-side without recovering moored hardware.

Several limitations remain. The evaluation uses static 2-second clips under controlled conditions, which does not fully capture continuous marine data streams. The highly compressed latent space is specialized for classification and does not support reconstruction of high-fidelity audio, as the bottleneck discards task-irrelevant acoustic information by design. The predefined 15-class subset means the representations are rigid with respect to out-of-distribution species or novel noise profiles. Additionally, while the bottleneck AE adds negligible inference overhead (Section III-C), the full pipeline still requires running the 12M-parameter Perch model at the edge; embedded deployment on ultra-low-power microcontrollers may require further model distillation before the framework can operate on the most resource-constrained buoy hardware.

Future work will extend the framework to continuous stream processing, explore lightweight neural vocoders for receiver-side audio reconstruction from compressed latent codes, and investigate open-set recognition and few-shot adaptation to allow deployed monitoring systems to generalize to novel species without hardware recovery.

ACKNOWLEDGMENT

The authors would like to thank the Watkins Marine Mammal Sound Database Team, Woods Hole Oceanographic Institution, and the New Bedford Whaling Museum for curating and providing the dataset used in this study.

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